CHAPTER 35

Missing Data - Complete Case Analysis

Missingness Mechanisms, MCAR/MAR/MNAR, CCA Decision Framework

CampusX — 100 Days of Machine Learning
Feature Engineering Series, Videos 31–40

35.1 Understanding Missing Data

Missing data is one of the most common problems in real-world machine learning. A value is "missing" when no information is recorded for a particular observation on a particular feature. In Pandas, missing values are represented as `NaN` (Not a Number) for numeric data and `None` for object data. In Scikit-learn, missing values are represented as `np.nan`.

35.1.1 Why Do Missing Values Occur?

- **Sensor failure:** A physical sensor malfunctioned during recording.
- **Human error:** A data-entry operator skipped a field.
- **Survey non-response:** A respondent left a question blank.
- **Data integration:** Merging datasets from different sources where some fields do not overlap.
- **Privacy restrictions:** Sensitive fields were deliberately withheld.
- **Conditional collection:** A follow-up question was only asked if the previous answer was "yes".

35.2 Missingness Mechanisms

Understanding *why* data is missing is crucial because it determines which handling methods are statistically valid. Rubin (1976) classified missingness into three mechanisms:

Mechanism	Abbreviation	Definition	Example
Missing Completely at Random	MCAR	Missingness is independent of both observed and unobserved data	Random equipment malfunction
Missing at Random	MAR	Missingness depends on observed data but not on the missing value itself	Younger patients skip income question more often
Missing Not at Random	MNAR	Missingness depends on the missing value itself	High-income earners refuse to report income

Table 35.1: Rubin's three missingness mechanisms.

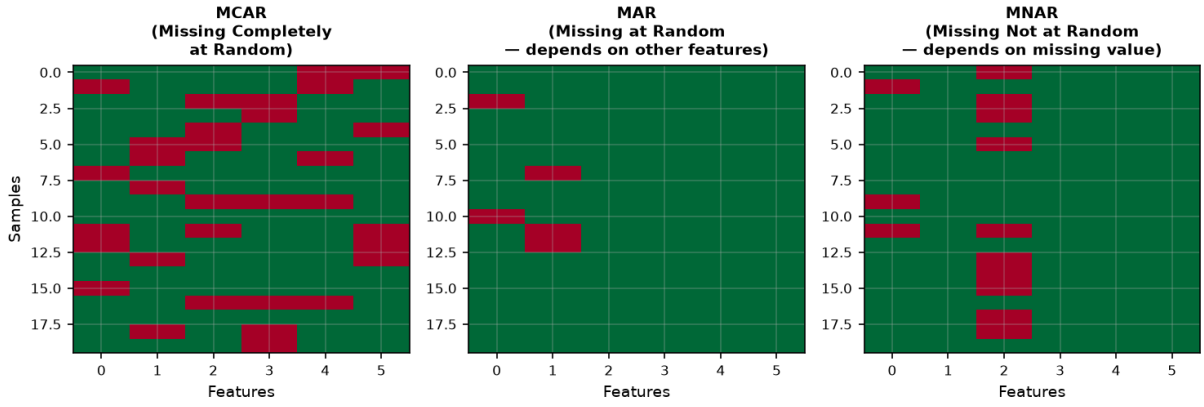


Figure 35.1: Visual representation of missing data patterns. Green cells are observed; red cells are missing. MCAR shows random scatter. MAR shows dependence on other observed features. MNAR shows dependence on the unobserved value itself (often harder to detect).

Why This Matters

If data is **MCAR**, removing incomplete rows (CCA) produces unbiased results. If data is **MAR** or **MNAR**, CCA can introduce bias because the missing pattern is informative. Understanding the mechanism guides the choice of method.

35.3 Complete Case Analysis (CCA)

Complete Case Analysis (also called listwise deletion) removes any row that contains at least one missing value. Only rows where every feature is observed are retained.

$$\text{CCA keeps row } i \iff \forall j:\ x_{ij} \neq \text{NaN} \tag{35.1}$$

35.3.1 Python Implementation

```
import pandas as pd
import numpy as np

# Create dataset with missing values
np.random.seed(42)
n = 200
df = pd.DataFrame({
'age': np.random.normal(35, 10, n),
'income': np.random.lognormal(10.5, 0.7, n),
'score': np.random.normal(70, 15, n),
'city': np.random.choice(['NYC', 'LA', 'Chicago', 'Houston'], n)
})

# Introduce missing values (MCAR - random)
for col in ['age', 'income', 'score']:
mask = np.random.random(n) < 0.15
df.loc[mask, col] = np.nan

print("=== MISSING DATA SUMMARY ===")
print(df.isnull().sum())
print(f"\nTotal rows: {len(df)}")
print(f"Rows with any missing: {df.isnull().any(axis=1).sum()}")
print(f"Percentage missing rows: {df.isnull().any(axis=1).mean()*100:.1f}%")

# Apply CCA
df_complete = df.dropna()
print(f"\n=== AFTER CCA ===")
print(f"Rows remaining: {len(df_complete)}")
print(f"Rows removed: {len(df) - len(df_complete)}")
print(f"Data loss: {(1 - len(df_complete)/len(df))*100:.1f}%")

# Compare statistics before and after
print(f"\n=== STATISTICAL COMPARISON ===")
print(f"{'Feature':<10} {'Full Mean':>10} {'CCA Mean':>10} {'Full Std':>10} {'CCA Std':>10}")
for col in ['age', 'income', 'score']:
print(f"{col:<10} {df[col].mean():>10.2f} {df_complete[col].mean():>10.2f} "
f"{df[col].std():>10.2f} {df_complete[col].std():>10.2f}")
```

Listing 35.1: Complete Case Analysis implementation.

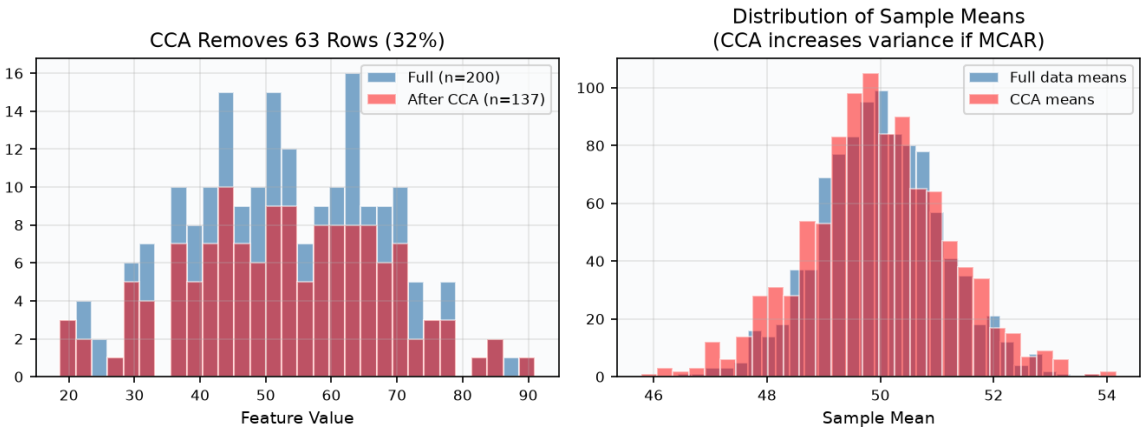


Figure 35.2: Left: CCA removes a substantial fraction of data. Right: Under MCAR, CCA preserves the distribution shape but reduces sample size, increasing the variance of parameter estimates.

35.4 When CCA is Appropriate

CCA is statistically valid when:

- The missingness mechanism is **MCAR** (missing completely at random).
- The proportion of missing data is **small** (typically < 5%).
- The dataset is **large enough** that losing some rows does not critically reduce statistical power.
- The missing values are not systematically related to the target variable.

35.5 When CCA Should be Avoided

- Missingness is **MAR** or **MNAR** — deletion introduces bias.
- The **percentage of missing data is high** (e.g., > 25%), leading to substantial information loss.
- The dataset is already **small** — every row matters.
- Missingness is **concentrated in important features** — dropping those rows removes valuable observations.
- The missing pattern is **systematic** — e.g., all patients who answered "yes" to a screening question have a follow-up value missing.

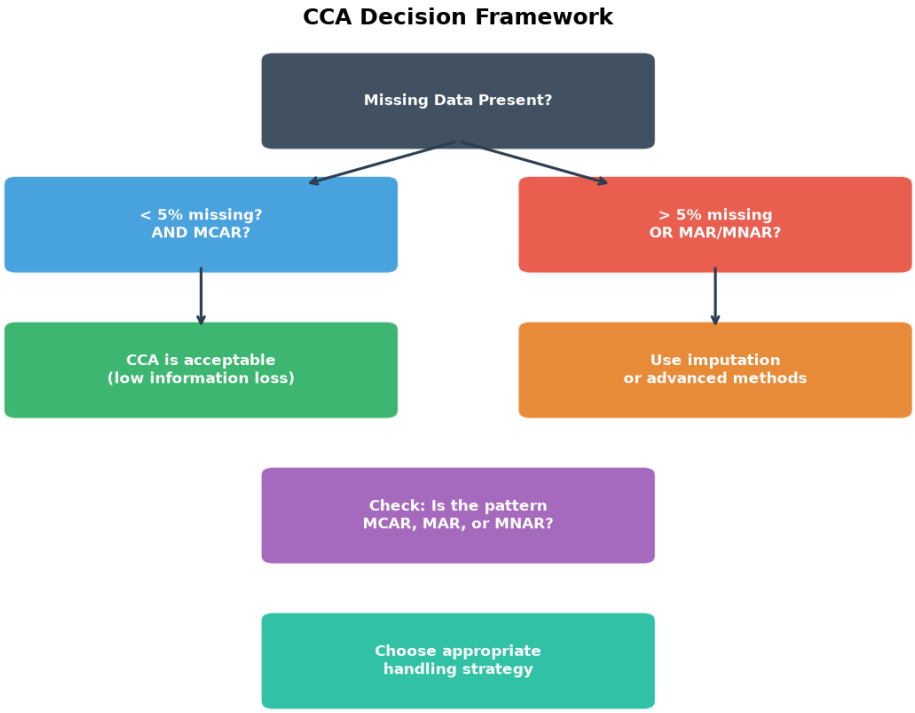


Figure 35.3: Decision framework for choosing between CCA and imputation.

35.6 Key Takeaways

> **★ Complete Case Analysis removes all rows containing any missing value. It is valid only when data is MCAR and the missing proportion is small. For MAR/MNAR or high missingness, imputation methods (Chapters 36–40) are preferred.**

35.7 Interview Questions

1. Explain the difference between MCAR, MAR, and MNAR with examples.
2. Under what conditions is Complete Case Analysis statistically valid?
3. What happens to the variance of parameter estimates when you apply CCA?
4. How would you test whether data is MCAR?
5. Why is CCA biased when data is MNAR?